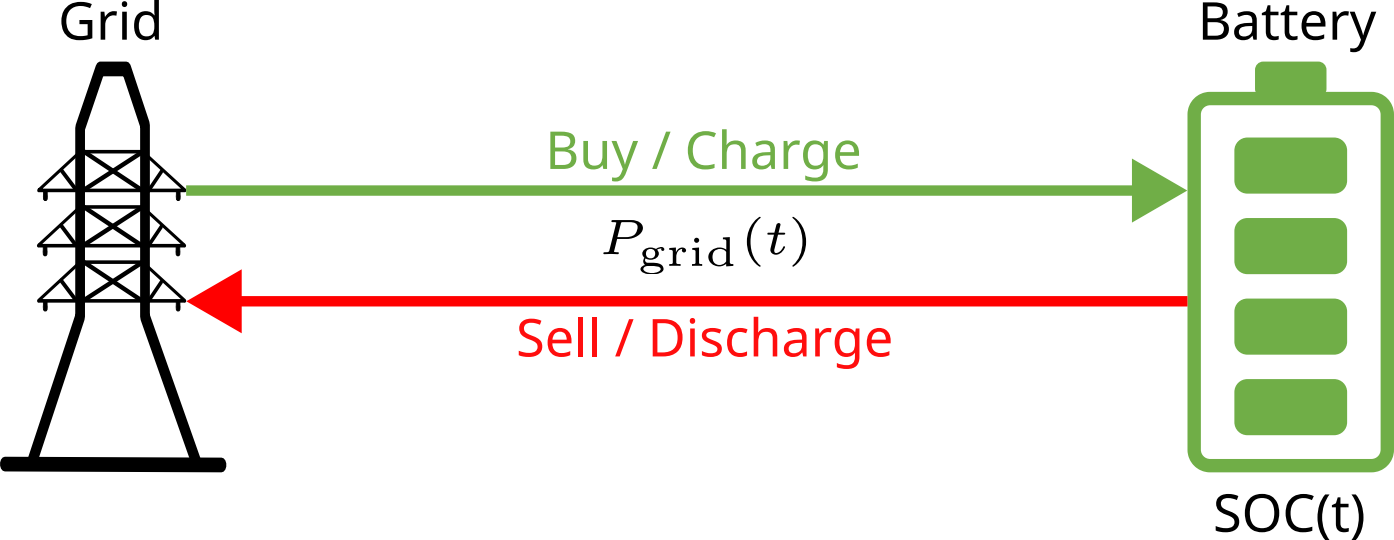
Grid
Battery
Buy / Charge
$P_{\mathrm{grid}}(t)$
Sell / Discharge
SOC(t)